Magnetism and age of the Porteau Pluton, southern Coast Belt, British Columbia: evidence for tilt and translation[1]

E. Irving, J. Baker, N. Wright, C.J. Yorath, R.J. Enkin, and D. York

Abstract: The Porteau Pluton is a variably foliated quartz diorite to granodiorite intrusion in the southern Coast Belt of the Canadian Cordillera (49.6°N, 123.2°W). $^{40}Ar/^{39}Ar$ ages are 95 ± 5 Ma from biotite and 101.5 ± 0.7 Ma from hornblende, which, together with an earlier U–Pb zircon age of 100 ± 2 Ma, indicate that the body was emplaced, uplifted, and cooled rapidly in mid-Cretaceous time. The rocks contain high coercive force (hard) remanent magnetizations with unblocking temperatures between 500 and 600°C, close to those of Ar in hornblende, indicating that remanence was acquired at or close to the hornblende plateau age. The hard remanence directions have an elongate distribution, in agreement with the predictions of M.E. Beck regarding magnetization acquired during tilting, uplift, and cooling of plutons. No part of the distribution agrees with the direction expected from observations from rocks of mid-Cretaceous age from cratonic North America. The elongate distribution defines the axis of tilt (347° east of north) but not its direction; tilt could have been down toward the east or down toward the west. The former yields an inclination that is 29.0 ± 4.9° shallower than expected from cratonic observations, corresponding to a displacement from the south of 3200 ± 500 km. The latter reconstruction yields an inclination that is anomalously shallow by 14.8 ± 3.9°, corresponding to a displacement from the south of 1600 ± 400 km, which is a minimum estimate. It is argued, therefore, that the Porteau Pluton has undergone both tilt and displacement from the south by distances substantially in excess of 1000 km.

Résumé : Le Pluton de Porteau est une intrusion dont la composition varie d'une diorite quartzique à une granodiorite, présentant une structure feuilletée variable, elle est localisée dans le Domaine côtier de la Cordillère canadienne (49,6°N, 123,2°W). Les âges $^{40}Ar/^{39}Ar$ sont 95 ± 5 Ma sur biotite et 101,5 ± 0,7 Ma sur hornblende, couplés à un âge U–Pb sur zircon de 100 ± 2 Ma déjà connu indiquent pour ce corps igné fut mis en place, soulevé et refroidit rapidement au Crétacé moyen. Les roches sont caractérisées par des aimantations rémanentes à forte coercivité (dures), et leurs températures de déblocage comprises entre 500 et 600°C, voisines de celles de l'Ar dans la hornblende, indiquant que la période d'acquisition de la rémanence correspond de près ou directement à l'âge plateau fourni par la hornblende. Les directions de la rémanence dure présentent une distribution allongée, en accord avec les prédictions formulées par M.E. Beck à propos d'une aimantation acquise durant des évènements plutoniques de basculement, soulèvement et refroidissement. Aucune portion de cette distribution n'est compatible avec la direction déduite des observations sur les roches d'âge crétacé moyen du craton de l'Amérique du Nord. La distribution allongée définit l'axe de basculement (347° est du nord) mais pas sa direction; deux hypothèses sont plausibles, un basculement avec abaissement vers l'est ou un basculement avec abaissement vers l'ouest. La première hypothèse fournit une inclinaison 29,0 ± 4,9° moins profonde que celle anticipée sur la base des observations cratoniques, indiquant un déplacement à partir du sud de 3200 ± 500 km. La deuxième hypothèse fournit une inclinaison de 14,8 ± 3,9°, anormalement peu profonde, correspondant à un déplacement en partant du sud de 1600 ± 400 km, ce qui est une estimation minimale. Par conséquent, nous proposons que le Pluton de Porteau fut basculé et déplacé à partir du sud d'une distance qui excède substantiellement 1000 km. [Traduit par la rédaction]

Received April 18, 1994. Accepted September 16, 1994.

E. Irving,[2] C.J. Yorath, and R.J. Enkin. Geological Survey of Canada, Pacific Geoscience Centre, P.O. Box 6000, Sidney, BC V8L 4B2, Canada.
J. Baker. 1110 Maple Road, Sidney, BC V8L 3X9, Canada.
N. Wright. Macdonald Dettwiler, 13800 Commerce Parkway, Richmond, BC V6V 2J3, Canada.
D. York. Department of Physics, Univerity of Toronto, Toronto, ON M5S 1A7, Canada.

[1] Geological Survey of Canada Contribution 29693.
[2] Corresponding author (e-mail: enkin@pgc.emr.ca).

Introduction

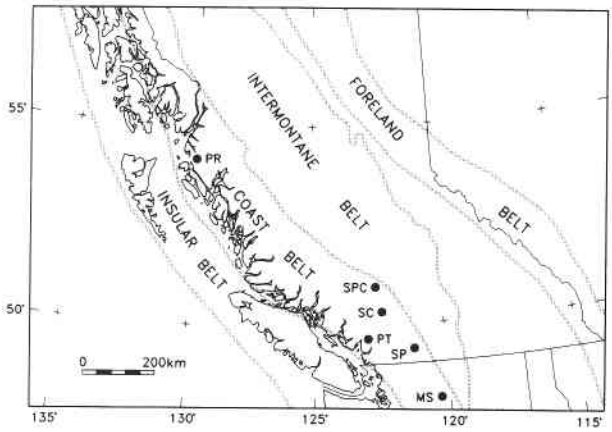
Over the past 20 years paleomagnetic studies have been conducted on mid-Cretaceous plutons from the Coast Plutonic Complex and Cascade Mountains of British Columbia and northern Washington, which together constitute the Coast Belt (Fig. 1). Their magnetization directions are discordant with those calculated from measurements made on mid-Cretaceous rocks collected from the North American craton. Some workers have argued that the discordances were caused largely by tilting (Symons 1974, 1977*a*, 1977*b*; Butler et al. 1989; Vandall 1993). Other workers have argued that the discordances imply displacement of the Coast Belt from the south by 2000–3000 km since mid-Cretaceous time (Beck and Noson 1972; Beck et al. 1981; Beck 1980, 1992; Irving and Wynne 1990; Irving et al. 1985; Ague and Brandon 1992).

This debate may be resolved by (1) obtaining paleomagnetic observations from coeval bedded sequences, which provide the means of correcting for tilt (Wynne et al. 1995, (2) using petrographic depth indicators to determine tilt of plutons (Irving and Archibald 1990; Ague and Brandon 1992), or (3) observing the manner in which directions of remanent magnetization are distributed within an intrusion. If a pluton is tilted at the same time as it is being uplifted and cooled, different parts of it would acquire a thermoremanent magnetization (TRM) at different stages during tilting. As a consequence, the directions of remanent magnetization would be distributed along a small circle centred on the horizontal axis of rotation (Beck 1991). Information about tilt, therefore, could, in principle, be obtained by observing and analysing such elongated distributions of directions of magnetization. In practice, small-circle distributions would be difficult to distinguish from great-circle distributions caused by processes other than simple tilting, by differential tilting, partial overprinting, or magnetic anisotropy, for example, and these possibilities need to be investigated.

In previous paleomagnetic studies of plutons, elongated distributions of directions have not been found. We describe such a distribution in the mid-Cretaceous Porteau Pluton of the southern Coast Belt (Fig. 1). From demagnetization, geochronometric, and petrographic studies, we estimate the time at which magnetization was acquired, and from studies of magnetic properties, we determine the reliability of the magnetization as a recorder of the paleofield. On these bases, we argue that tilting is the most likely cause of the elongate distribution. We show that the tilt required does not bring the magnetic inclination into agreement with that expected from cratonic reference field, the implication being that, since its emplacement, the pluton has moved a considerable distance from the south relative to cratonic North America.

Symons (1973) obtained paleomagnetic results from the Porteau Pluton, as part of an extensive study of several southern Coast Belt plutons. He argued that their magnetization was consistent with that expected from cratonic North America, implying no motion of the Coast Belt since mid-Cretaceous time. However, Beck (1976) noted that the inclination of the magnetization directions observed by Symons actually were systematically shallower than expected, implying transport of the Coast Belt from the south relative to the craton. Irving et al. (1985) confirmed Beck's inference by observing markedly shallow inclinations at one site (I8, Fig. 2) in the Porteau Pluton, an inclination that was in good agreement with their more extensive study of the Spuzzum Pluton (Fig. 1) of the eastern Coast Belt. The present study was undertaken to determine the reliability of this agreement, but as work progressed, and the elongated distribution of directions came to light, it became clear that the cause of this distribution needed to be established before a comparison with Spuzzum could be made.

Fig. 1. Morphotectonic belts of Canadian Cordillera and locations of the mid-Cretaceous rocks that have been studied paleomagnetically. Plutons: Ms, Mt. Stuart; PT, Porteau; PR, Prince Rupert; SC, Spetch Creek; SP, Spuzzum. SPC, Silverquick and Powell Creek formations.



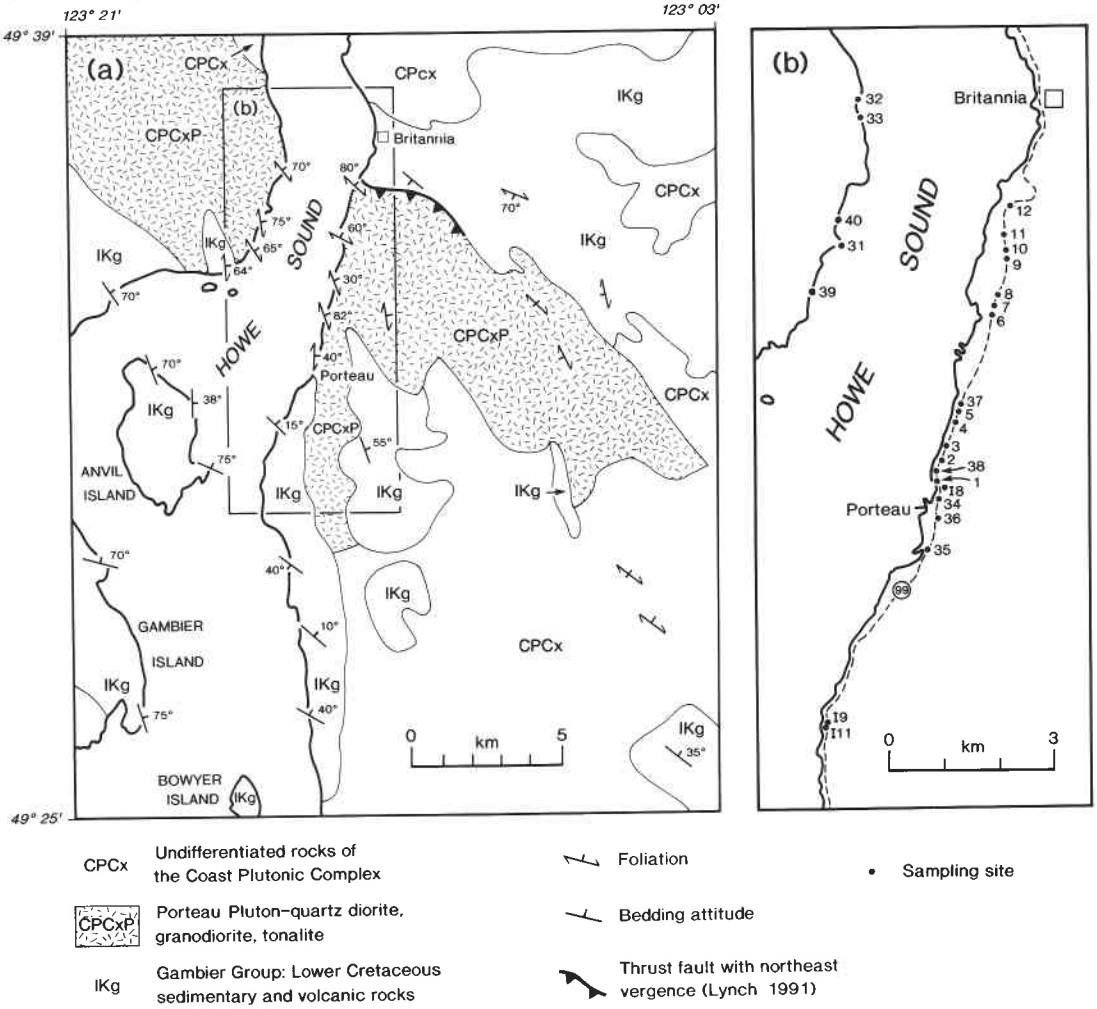
Geology and sampling

The Porteau Pluton in the southern Coast Belt of British Columbia is situated in the central Howe Sound region (Fig. 2). It is a leucocratic, coarse-grained, variably foliated, hornblende-rich quartz diorite–granodiorite (J.A. Roddick 1987). The pluton intrudes middle Albian strata of the Gambier Group (J.A. Roddick 1987). The body is aligned northwesterly and its outcrop is coincident with a northwesterly trending aeromagnetic low that crosses Howe Sound without apparent disruption. It is therefore reasonable to assume that the Porteau Pluton is a coherent body unbroken by very large faults.

The rocks generally are foliated, with alternating irregular light (quartz–feldspar) and dark (hornblende–biotite) bands. Foliation is commonly moderate to strong, but locally is weak. Along the southeast side of Howe Sound, foliations strike generally north to northwesterly and dip moderately to steeply down to the northeast. On the western shore, foliations have the same trend and are steep to vertical. On the southeast side of the sound, the northern boundary between the Porteau Pluton and the Gambier Group is interpreted by Lynch (1991) to be a thrust fault, which, based upon fabrics observed in a mylonite 50 m thick, is considered to represent northeasterly transport of the pluton over Gambier Group volcanic rocks.

Oriented cores, generally 5 per site, were obtained from 23 sites (Fig. 2). Orientation was by sun and magnetic compasses. Generally, two specimens were cut from each core.

Fig. 2. (*a*) Regional geology of Howe Sound region, modified from J.A. Roddick and Woodsworth (1979) and Lynch (1991). (*b*) Distribution of sampling sites (Table 1) enclosed by box in (*a*).



Geochronology

Zircon from site I8 provided a U–Pb date of 110 ± 2 Ma (R.L. Armstrong, personal communication to G.J. Woodsworth, 1983, see Irving et al. 1985, p. 587). $^{40}Ar/^{39}Ar$ data were obtained with the University of Toronto's argon-ion laser system (Layer et al. 1987), using procedures described by Wright et al. (1990). The system consists of a Spectra-Physics 171-18 (18 W) continuous argon-ion laser for sample heating, a Barnes Engineering RM-2A infrared microscope for determining sample temperature, and a VG-1200S mass spectrometer for measuring isotopes. Standards were single grains of hornblende Hb3gr (Zartman 1964) with an assumed age of 1071 Ma (J.C.M. Roddick 1983; Turner et al. 1971).

Thirty-seven heating runs on single grains from hornblende separates from site 31 were made. Individual grains often contained insufficient gas to give multiple-step age spectra. However, three to six grains, when heated together, produced reasonable age spectra. Ten multiple-step and 25 single-step experiments involving 72 grains were successfully completed. The former produced plateaux throughout the entire release, indicative of rapid cooling. Typical spectra are shown in Fig. 3. Summing the results, as if all 72 grains had been step-heated together, yields an integrated age of 101.7 ± 0.7 Ma.

Five biotite grains from sites 3, 5, and 8 (Fig. 2) were studied, employing three multiple-step and two single-step heating runs. Site 3 yielded no plateau. Within 1σ errors, the spectra for sites 5 and 8 (two shown in Fig. 4) form crude plateaux. The average of five grains is 95 ± 5 Ma, the large error reflecting the discordant nature of the spectra.

The U–Pb age indicates intrusion close to 100 Ma ago. Cooling through the argon blocking temperatures in hornblende and biotite apparently required no more than a few million years. Assuming blocking temperatures of 500–600°C for hornblende, and 250–350°C for biotite (Harrison 1981; York 1984; Harrison et al. 1985), a cooling rate of about 60°C/Ma can be calculated for the interval 102 Ma to 95 Ma. The flat hornblende age spectra show no evidence of subsequent reheating that might have caused remagnetization.

Paleomagnetism

Measurements were made on automated Schonstedt spinner magnetometers. Stepwise alternating magnetic field (AF) demagnetization was carried out using a Schonstedt GSD-5

Fig. 3. Hornblende age spectra, site 31. The 1σ error level shown for each step does not include errors due to the uncertainties in standards. I.A., integrated age.

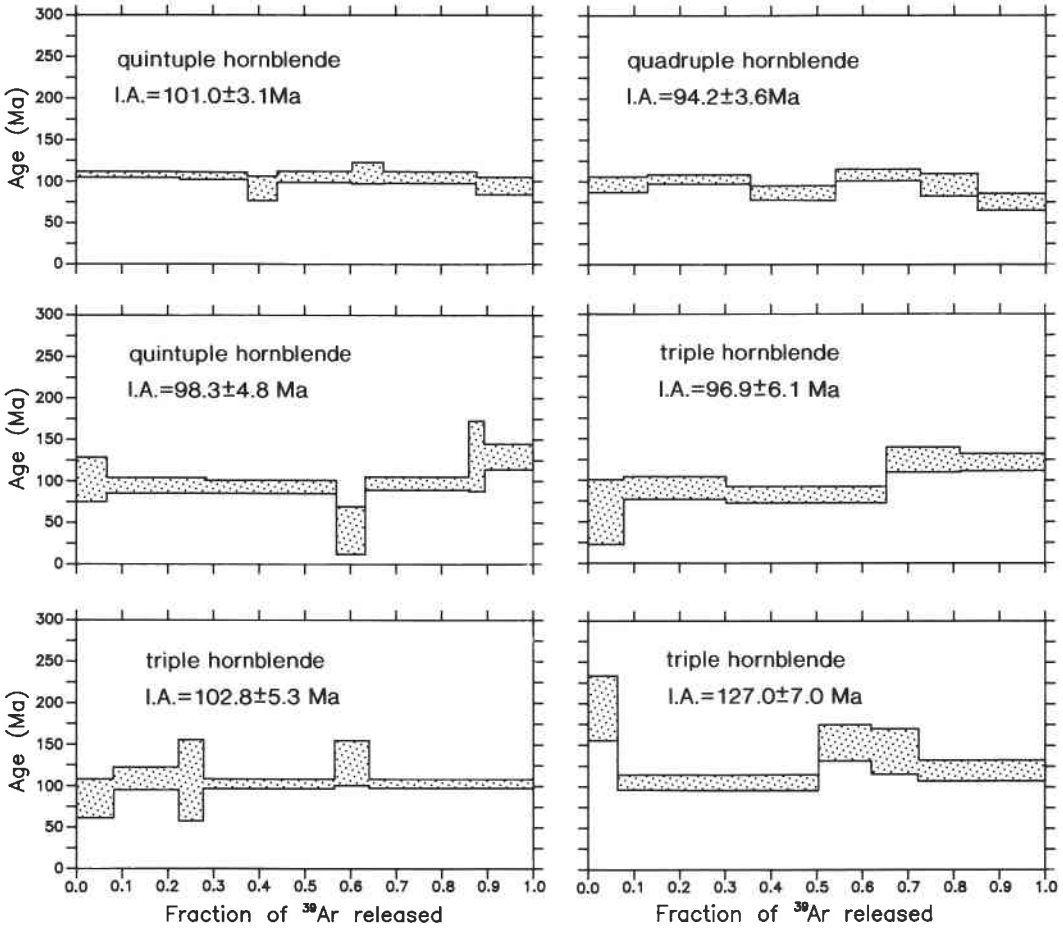
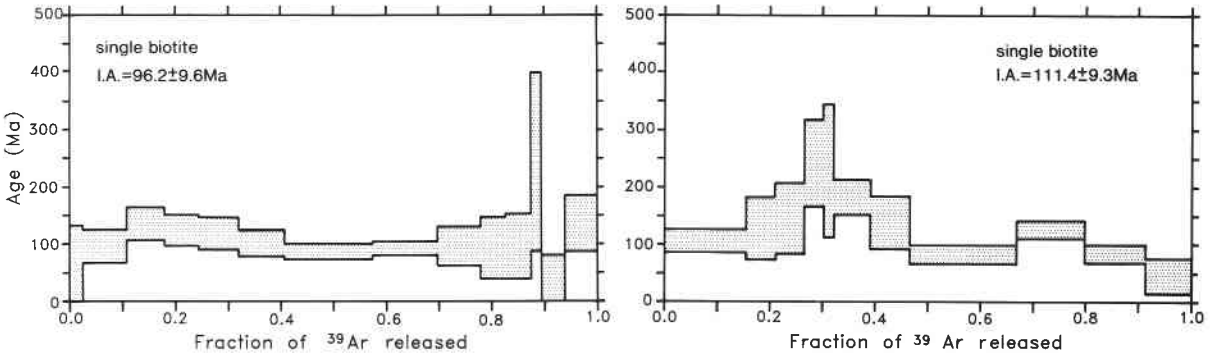


Fig. 4. Biotite age spectra, site I8.

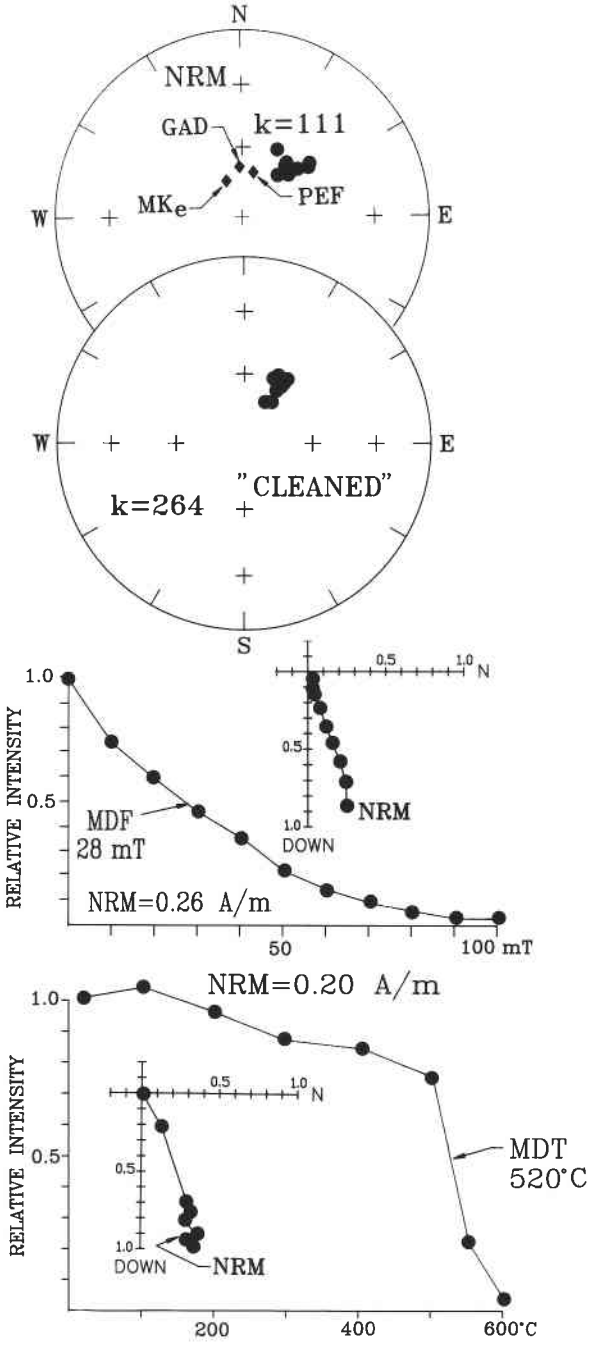


instrument. For thermal demagnetization a Schonstedt TSD1 furnace was used.

The natural remanent magnetizations (NRM) fall into three categories. **X** magnetizations have a single, high coercive force (hard) magnetization, occasionally with small magnetically soft overprints directed along the present field (interpreted as Brunhes overprints). Their unblocking temperatures (T_{UB}) range from 500 to 600°C, and the median destructive fields (MDF, the AF required to reduce the NRM by 50%) average 33 mT, always exceeding 18 mT. **Y** magnetizations have a hard component, very similar in character to **X**, together with large, soft Brunhes overprints of variable magnitude; the former typically constitutes about 30% of the magnetization present. T_{UB}'s of **Y** range from 20 to 600°C, and MDF's average 12 mT, always exceeding 10 mT. **Z** magnetizations have T_{UB}'s distributed between room temperature and 500°C and MDF's that average 8 mT, never exceeding 10 mT, and are interpreted as Brunhes overprints.

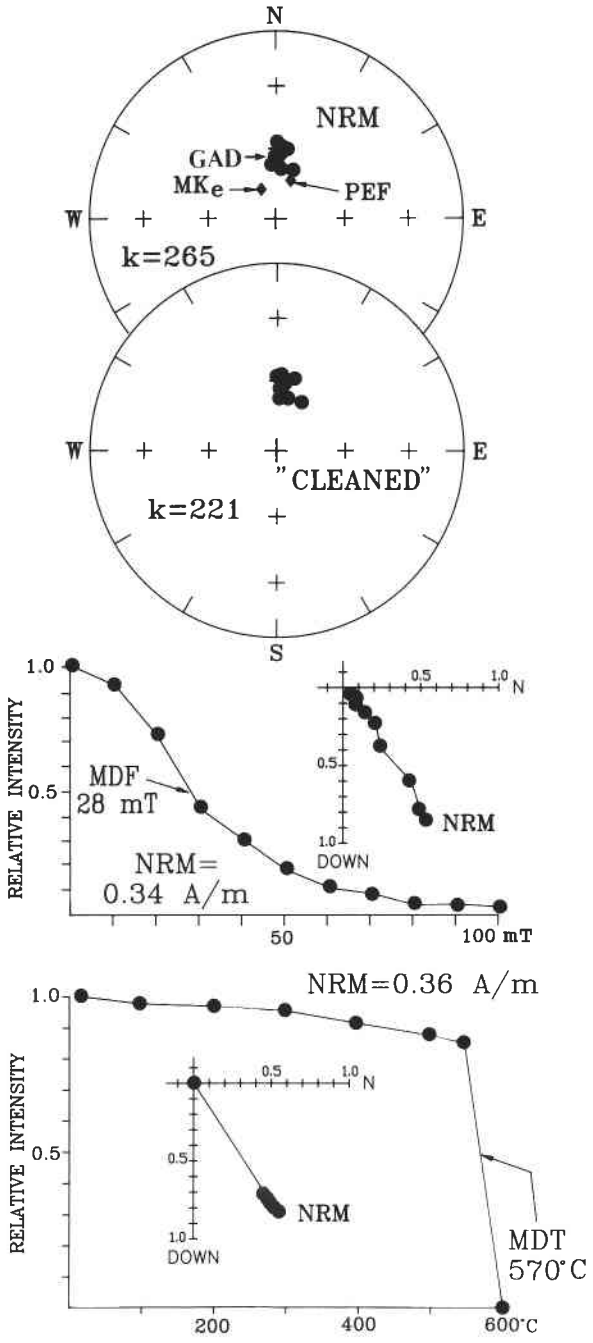
An example of **X** magnetization is given in Fig. 5. NRM directions are well-grouped and directed towards the northeast with intermediate downward inclinations. The thermal decay curve is square-shouldered. Orthogonal plots show

Fig. 5. X magnetization, northeast directions, site 38. Top: the directions of natural remanent magnetization (NRM) in nine specimens from five cores compared with direction of the present Earth's magnetic field (PEF), the geocentric axial dipole field (GAD), and the expected mid-Cretaceous paleofield (MK_e). Middle: the directions of hard magnetization. Bottom: AF and thermal decay curves and orthogonal plots for individual specimens. MDF and MDT are the median destructive field (AF) and temperature respectively.



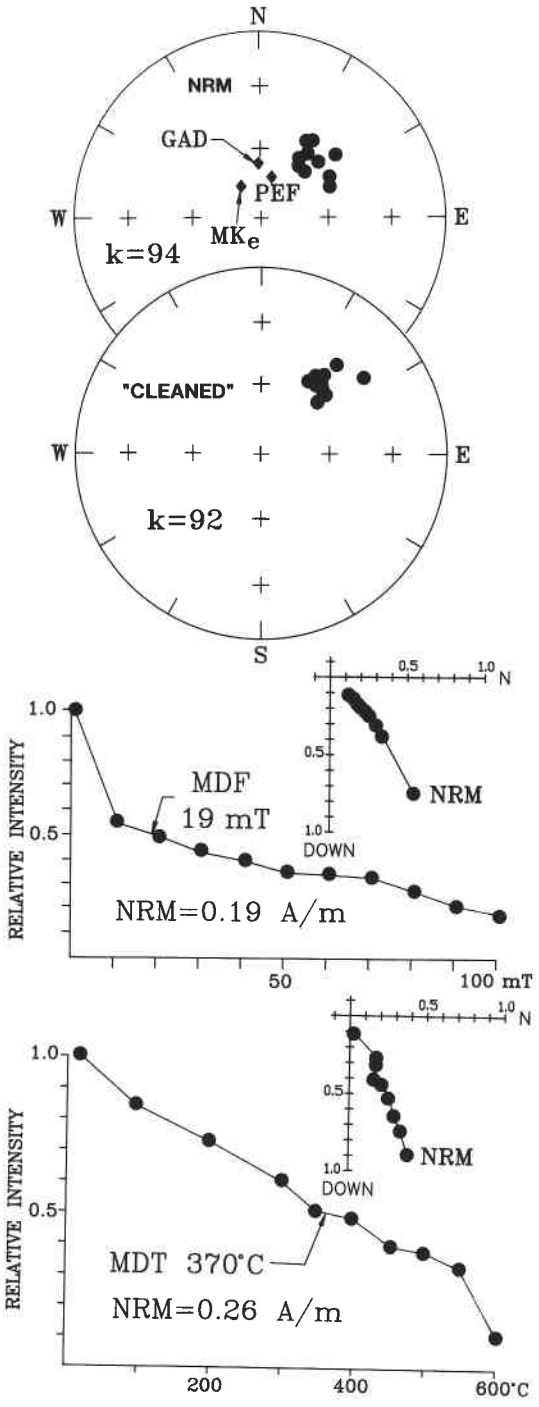
linear decay to the origin at AF's above 10 mT. After cleaning, dispersion decreases. The second example generally has similar characteristics, but the directions are northerly (Fig. 6). Site 38 differs from the present geocentric axial dipole field

Fig. 6. X magnetization, northerly directions, site 31. See caption Fig. 5.



(which is close to the average mid-Tertiary and Quarternary field) but site 31 does not. An example of **Y** magnetization is given in Fig. 7. AF demagnetization initially causes a rapid decrease in intensity, following which, directions remain little changed up to 100 mT. Both AF and thermal demagnetization yield good end points. Orthogonal plots show linear decay at higher AF and temperatures. After cleaning, the agreement among specimens is good. There is a small general shift away from the present Earth's field as large Brunhes overprints are removed. An example of **Z** magnetization is given in Fig. 8. The intensity falls very rapidly during AF demagnetization, and T_{UB}'s are widely

Fig. 7. Y magnetization, site 2. See caption Fig. 5.



distributed (Fig. 8). Directions are close to the present Earth's field.

Directions of magnetization for each specimen were determined using principal-component analysis. Lines are drawn through a minimum of four points and the origin. To obtain **X** and **Y** magnetizations, lines were drawn employing observations in the range 10−80 mT or 400−600°C. To obtain **Z**, the ranges were 0−20 mT or 0−300°C. Mean directions for sites are given in Table 1 and summarized in Table 2. Within each site, **X** and **Y** magnetizations are very well grouped; Fisher precisions for **X** sites are typically 100−200

Fig. 8. Z magnetization, site 8. See caption Fig. 5.

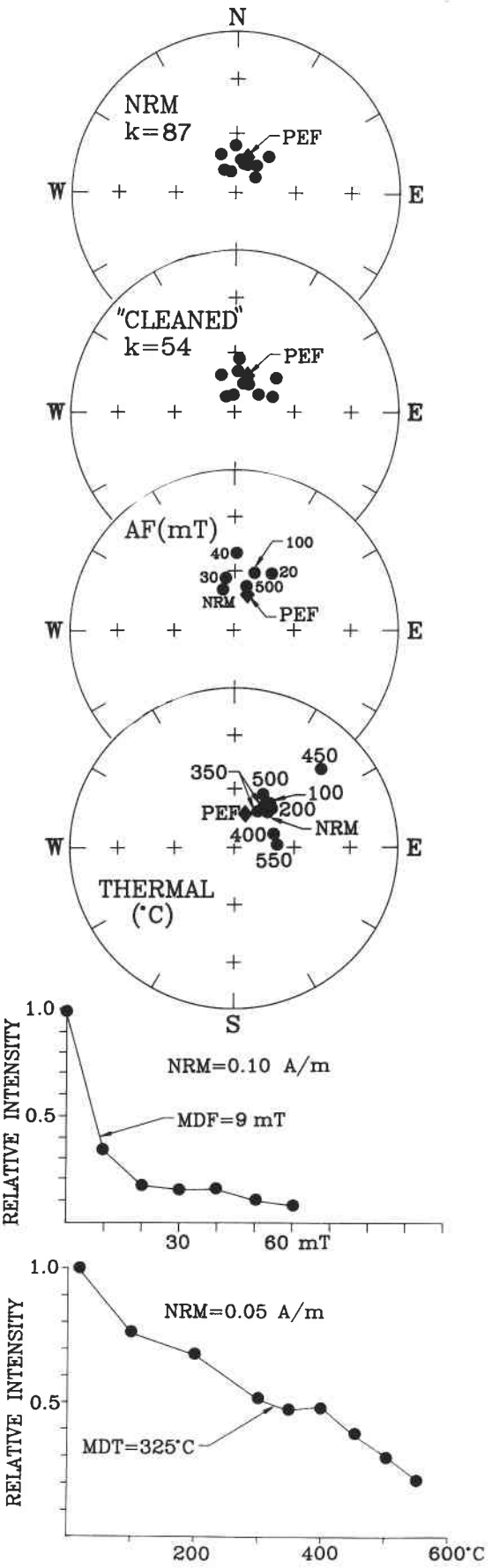


Table 1. Paleomagnetic data by sites.

Site	UTM (E,N)	M (A/m)	n	s	D (°)	I (°)	k	α_{95} (°)	Rock
01(Y)	483300,5489800	0.356	4	6	41	51	257	4	Mafic xenolith
02(Y)	483300,5490100	0.218	5	10	43	47	85	5	Mafic lense
03(X)	483400,5490500	0.217	4	7	27	53	381	3	di
04(X)	483600,5491000	0.041	2	6	36	57	278	4	Mafic lense
04(Y)	—	0.115	3	7	25	65	78	8	Mafic lense
04(all)	—	0.081	5	12	31	61	87	5	—
05(X)	483700,5491200	0.768	3	6	15	64	403	3	di, gd
06(X)	484300,5492900	0.618	5	8	348	62	129	5	Mafic lense
07(Z)	484300,5493100	0.071	5	9	355	82	55	7	di, gab
08(Z)	484400,5493300	0.118	5	10	19	77	47	7	di, gab
09(Y)	484600,5493900	0.051	5	12	356	61	103	5	Mafic xenoliths
10(Y)	484600,5494000	0.043	5	10	342	64	94	5	di, xenoliths
11(Y)	484500,5494400	0.232	5	10	331	72	33	9	Dike
12(Z)	484600,5494800	0.227	5	8	197	83	42	9	di
31(X)	481500,5494100	0.359	5	10	3	60	449	3	di, xenolith
32(X)	481900,5496800	0.797	3	6	356	58	136	6	qu.di
32(Z)	—	0.219	2	4	0	90	79	10	qu.di
33(Y)	481900,5496400	0.530	5	9	0	69	87	6	gd, xenolith
34(Z)	483300,5489500	0.360	5	5	51	58	52	13	di, xenolith
35(Y)	483100,5488600	0.878	5	10	30	47	68	6	qu.di
36(Z)	483330,5489200	0.618	4	6	36	49	146	6	qu.di
37(X)	483700,5491300	0.656	6	11	17	66	378	3	qu.di
38(X)	483400,5490300	0.281	5	7	30	62	234	4	qu.di
39(Z)	481100,5493300	0.288	5	5	65	80	27	15	qu.di
40(Y)	481500,5494500	0.195	6	9	21	59	70	6	gd, xenolith
I8(X)	483400,5489700	0.110	5	10	22	52	142	4	tonalite
I9	481300,5485300	0.002	5	10	360	65	117	5	argillite
I11	481300,5483000	0.004	2	4	354	58	53	13	argillite

NOTES: M is magnetization intensity; n, s are number of cores and specimens, respectively; D, I are declination and inclination, respectively, of the mean direction giving specimens (s) unit weight; k is Fisher's estimate of precision; and a_{95} ($P = 0.05$) is radius of the circle of confidence. Rock types: di, diorite; gab, gabbro; qu.di, quartz diorite; gd, granodiorite. Sites I8, I9, and I11 site are from Irving et al. (1985) with additional information. Data from dike-like body site 11 are added for completeness.

or higher, and for **Y** sites 80–100 (Table 1). **Z** sites have lower precision, typically about 50. **Z** magnetizations are of low stability, and the directions group about the present field (Table 2). **X** and **Y** magnetizations have an elongated distribution, extending about 40° of arc in an east–west sense. Their directions differ from the present field and the expected mid-Cretaceous field (Fig. 9).

The mean direction obtained by Symons (1973) falls within this distribution (Table 2). His data have been obtained from several bodies (including Porteau Pluton) distributed over a wider area and so are not strictly comparable with ours. It is notable, however, that his mean inclination, like ours, is shallower than expected (MK_e) from cratonic information. Directions observed by Irving et al. (1985) from the Gambier Group collected to the south of the Porteau Pluton (Table 1; Fig. 2) also fall within this distribution (Table 2).

Magnetic properties

Isothermal remanent magnetization

Steady magnetic fields were applied to test specimens from each site and the isothermal remanent magnetization (IRM) acquired then measured. **Y** and **Z** specimens usually become saturated (saturation IRM or IRM_s) in <100 mT, typical for magnetite-bearing rocks. This is insufficient for most **X** specimens, which commonly require 200 mT or more to saturate, indicating the presence of fine-grained single-domain magnetite. The MDF of IRM_s averages 16 mT for **Z**, 20 mT for **Y**, and 24 mT for **X** specimens, showing the systematic increase in "hardness" through the categories established on the basis of NRM properties.

Lowrie–Fuller tests

Normalized AF demagnetization curves of IRM_s and NRM are compared in Fig. 10. In **X** specimens, the normalized NRM demagnetization curves always lie well above those for IRM_s, indicating that the magnetization probably is carried by single-domain grains. This also is true for **Y** magnetization, but the difference is less marked. For **Z** magnetization the two curves are similar. There is a clear distinction between **X** and **Z** specimens. **Y** has intermediate properties.

Long-term heating experiments

Pairs of specimens from the same site were demagnetized in

Table 2. Summary of paleomagnetic data.

Magnetism	N	D_m (°)	I_m (°)	k	α_{95} (°)
X	9	15.5	60.1	75	6.0
Y	9	23.0	58.5	33	9.0
X + Y	18	17.4	56.8	21	7.5
Z	6	42.7	81.3	35	11.5
Sy	17	350.8	65.2	18	7.9
MK_e	—	333.4	74.9	—	2.8

NOTES: The mean location is 49.6°N, 123.2°W. N is the number of sites; D_m and I_m are the declination and inclination, respectively, of mean direction; k is the precision, and α_{95} is the radius of the circle of confidence. The data are from 22 sites; site 4 has both **Y** and **Z** magnetization, and site 32, **X** and **Z**. The paleopole corresponding to the **X** magnetization is 77.0°N, 0.9°W, $K = 35$, $A_{95} = 8.8°$. Because of the elongated nature of the distribution of **X** and **Y** directions, their statistics must be used with caution; they are given here for the sake of completeness. Sy is the result of Symons (1973) from 17 sites in Mesozoic plutons distributed from Horseshoe Bay north to the Squamish River valley and includes six sites from the Porteau Pluton. MK_e is the expected mid-Cretaceous field calculated from the reference paleopole (72.2°N, 194.1°E, $K = 352$, $a_{95} = 3.7°$) derived from five studies summarized by van Fossen and Kent (1992).

10 mT to remove components of recent origin. They were then placed in a furnace and heated in the Earth's field to 300°C for increasing intervals of time, cooled to room temperature, and the vertical component of magnetization measured. One specimen of each pair was placed upright in the furnace, the other inverted, so by taking sums and differences of pairs, the decay of NRM and buildup of thermally enhanced viscous remanent magnetization (VRM) could be determined (Fig. 11). There was little or no decay of NRM, but VRM developed quickly, and increased steadily. In **Z** specimens, the VRM acquired was about twice as strong as NRM, whereas in **X** specimens it was one or two orders of magnitude less. There is a very clear distinction between **X** specimens and **Z** specimens.

Magnetic anisotropy

The rocks show varying and substantial degrees of anisotropy of magnetic susceptibility (AMS), which probably is caused in part by the alignment of hornblende and reflects the foliation visible in outcrop (Table 3). The bulk susceptibility and anisotropy are substantially less in **X** than in **Y** and **Z** specimens. To determine the effect that foliation might have had on the directions of magnetization, the specimens used for AMS studies were given an IRM_s by applying a field of 200 mT. The field was applied at large angles to the direction of maximum susceptibility. The IRM_s was then demagnetized (Fig. 12).

The IRM_s directions of **X** specimens were commonly parallel to the applied field, and remained so during AF demagnetization. If they were not, they soon become parallel after demagnetization in low AF; IRM remained accurately parallel to the applied field in the coercive force range >10 mT (Fig. 12), so there is no anisotropy of the high coercive force remanent magnetization. Hence foliation in

Fig. 9. Site mean direction for **X** and **Y** magnetizations. G, E, and F explained in text. MK_e expected direction from cratonic observations (Table 2).

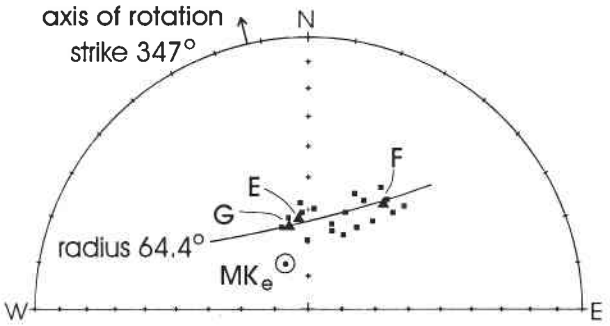
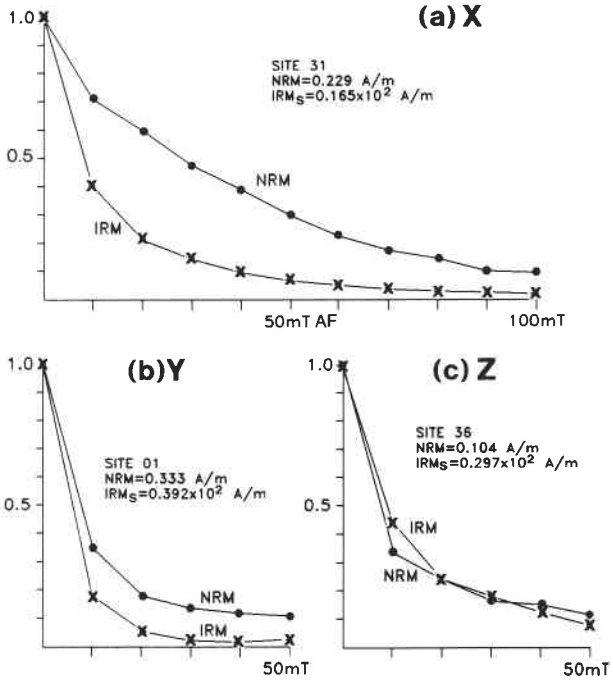


Fig. 10. Demagnetization of IRM_s and NRM normalized for comparison (Lowrie and Fuller 1971), for (*a*) **X**, (*b*) **Y**, and (*c*) **Z** specimens.



X specimens has negligible effect on hard remanence directions, although the rock is visibly foliated and AMS is significant (Table 3). We conclude that the ensemble of fine-grained magnetite grains, which carries the hard **X** magnetization, accurately records the paleofield. The IRM_s direction of **Y** specimens are never parallel to the applied field, occasionally being 20° from it. During AF demagnetization, the directions migrate towards but do not reach the applied field (site 40, Fig. 12). Ultimate departures from the applied field are ~5°. Demagnetization of IRMs of **Z** specimens is generally similar to that of **Y** specimens in this regard.

The experiments do not, of course, show that the high T_{UB} high coercive force component of **Y** NRM is anisotropic. It could very well be isotropic, as in **X** specimens. These experiments do show, however, that the effect of foliation on NRM In **Y** specimens is small—far too small to cause the spread of directions observed (Fig. 9).

Fig. 11. Buildup of viscous remanent magnetization (VRM) and retention of natural remanent magnetization (NRM) during long-term heating of **X** (site 31) and **Z** (sites 39 and 7) specimens.

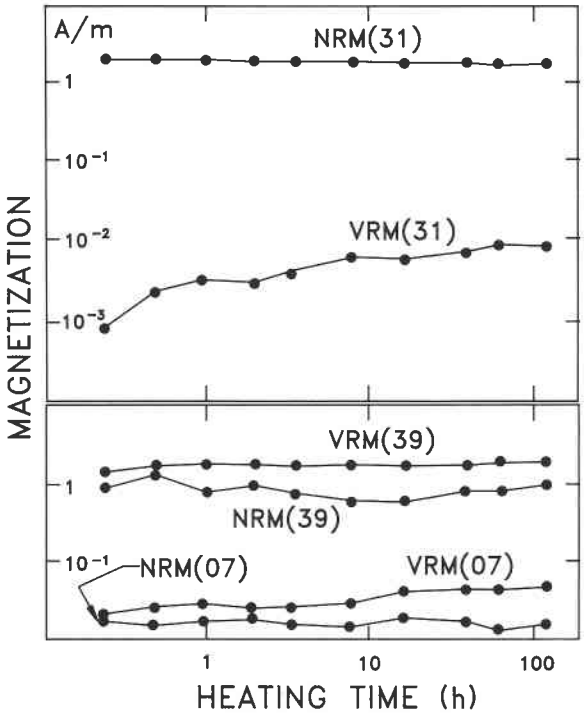


Table 3. Anisotropy of magnetic susceptibility.

Measures	**X** (s = 9)	**Y** (s = 12)	**Z** (s = 10)
$P = K_1/K_3$	1.19±0.07	1.43±0.25	1.34±0.14
$H = (K_1 - K_3)/K$	0.17±0.05	0.34±0.16	0.28±0.10
$L = K_1/K_2$	1.09±0.05	1.17±0.07	1.12±0.07
$F = K_2/K_3$	1.11±0.03	1.23±0.14	1.20±0.07
χ ($\times 10^{-3}$)	0.55±0.28	2.06±2.22	1.34±0.85

NOTES: P is the degree of anisotropy; H is total anisotropy; L is lineation; F is foliation; χ is the magnetic susceptibility in cgs units per cubic centimetre; χ (mks) = $4\pi\chi$ (cgs). Standard deviations of individual observations quoted. s is the number of specimens studied.

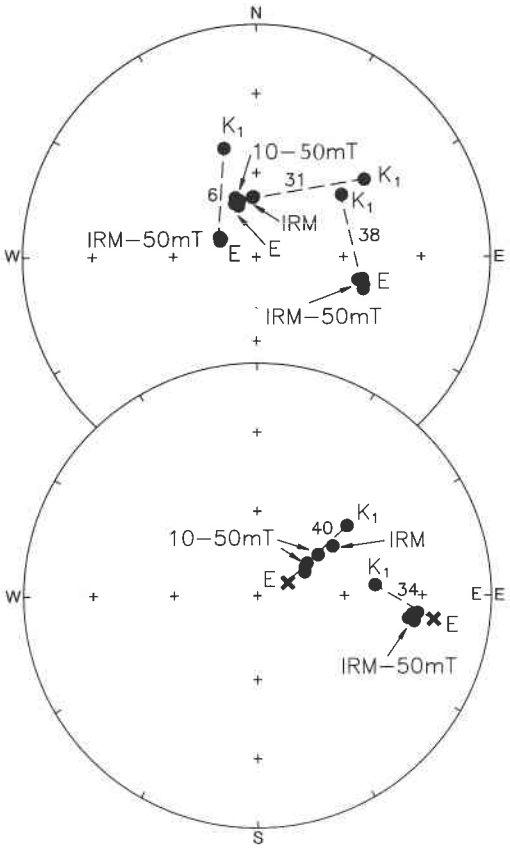
Carriers of remanence

In hand specimen, all rocks break with clean fracture. In polished thin sections, kaolinized feldspar occurred in one specimen, otherwise the rocks are fresh, with no evidence of low-temperature alteration.

Z specimens preserve an igneous mineralogy with little or no evidence of deuteric alteration. The rocks are strongly foliated. Magnetite grains range from >100 μm down to about 3 μm, and are commonly arranged along fabric layers (Fig. 13*a*). The fabric-controlled distribution accounts for the anisotropy of IRM (Fig. 12). The presence of large, presumably multidomain, grains explains the low coercive force, and low T_{UB}.

X specimens contain numerous, small magnetite grains (<5 μm), exsolved from large hornblende and to a lesser extent from plagioclase grains (Fig. 13*c*). The magnetite grains are sometimes equant, sometimes elongated. There is

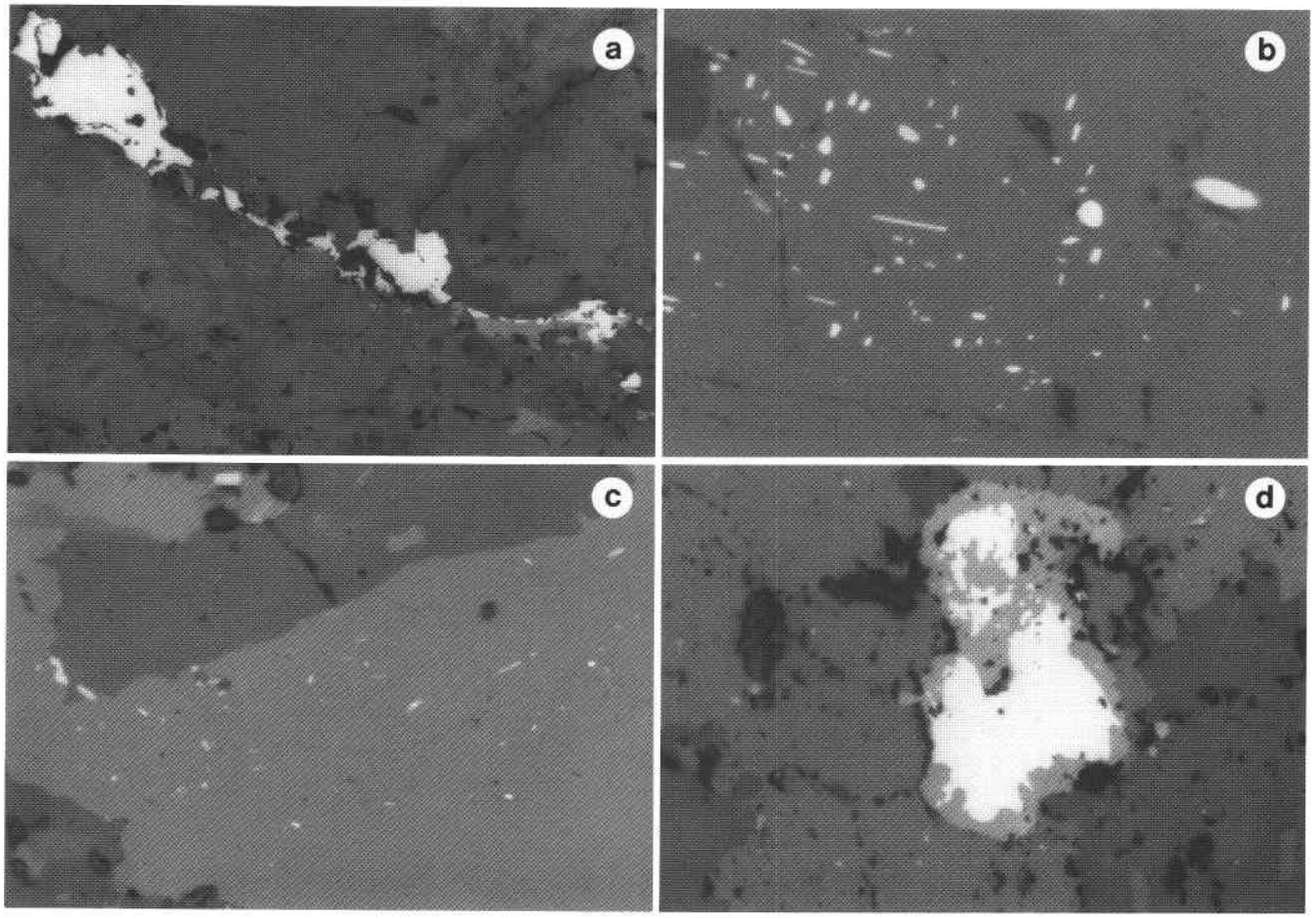
Fig. 12. Direction of isothermal remanent magnetization (IRM acquired in 200 mT) and maximum susceptibility (K_i) compared in **X** (sites 6, 31, 38), **Y** (site 40), and **Z** (site 34) specimens. E is applied field direction, denoted by cross where clustering allows. Directions of initial IRM, and after progressive AF demagnetization in fields indicated, are shown.



little or no preferred alignment within individual silicate grains, and none between grains in the same thin section. There also are large (typically 50–100 μm) grains of magnetite enclosed within or situated adjacent to much larger hornblende grains (Fig. 13*d*). They are surrounded by haloes of calcite, clear evidence of deuteric alteration. Margins of these large magnetite grains are highly convoluted, and commonly there are tiny magnetite grains within the carbonate haloes. It seems probable that the high discrete T_{UB}, high coercive force, single-domain magnetization of **X** specimens is carried by very fine-grained magnetite disseminated in the silicates, and presumable exsolved from them during deuteric alteration. Crystal structure of silicates might be expected to influence the shape of exsolved magnetite, thereby imposing a degree of magnetic anisotropy on the rock as a whole if the silicates themselves are preferentially aligned. The absence of preferred alignment of iron minerals across polished section, and the observed isotropy of hard remanence (Fig. 12), shows that such effects are negligible when averaged over a paleomagnetic specimen (11 cm^3). The role of calcite-rimmed magnetite grains is less clear, but their jagged contours make it possible that they, too, carry a stable isotropic magnetization.

Y specimens contain the iron minerals present in both **Z**

Fig. 13. Photomicrographs of polished sections taken in plane-polarized reflect light. (*a*) Site 18, **Y** magnetization, large magnetite grains, some having the form of intergrown euhedra, arranged in bands parallel to field of view foliation. (*b*) Site 18, **Y** magnetization, small magnetite inclusions in horneblende. (*c*) Site 31, **X** magnetization, small magnetite inclusions in host plagioclase (?). (*d*) Site 31, **X** magnetization, large irregular magnetite grain enclosed in carbonate. The opaque materials of **Z** specimens are generally very similar to (*a*). Horizontal dimension of fields of view are 600 μm (*a*), 125 μm (*b*, *c*), 250 μm (*d*).

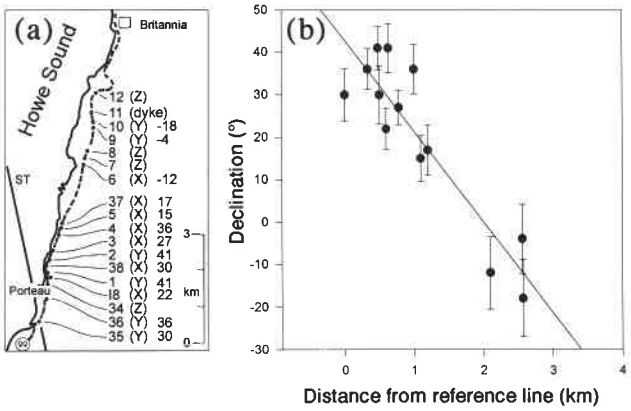


and **X** specimens. There are large magnetite grains aligned along the foliation, characteristic of **Z** specimens, and very small magnetite grains disseminated in silicates characteristic of **X** specimens (Figs. 13*a*, 13*b*). Large, irregularly margined magnetites with carbonate haloes also are present, as in **X** specimens. **Y** specimens show substantial foliation, as do **Z** specimens. Thus it is evident why **Y** specimens show both **X** and **Z** magnetic properties; the high T_{UB}, hard components probably are carried by fine-grained magnetite exsolved deuterically from silicates, and the large Brunhes overprints and anistropy of IRM is carried by the large fabric-oriented magnetite grains. Presumably, it is these large grains that also are responsible for the large bulk susceptibilities of **Y** and **Z** specimens (Table 3).

Age and origin of magnetization

X and **Y** magnetizations, freed of Brunhes overprints, have an elongate distribution ranging from northerly to northeasterly (Fig. 9). They have high coercive force and high discrete T_{UB}'s between 500 and 600°C, a range that includes the blocking temperature of ^{40}Ar in hornblende. IRM acquisition and Lowrie–Fuller tests indicate that in **X** specimens, single-domain magnetite is the predominant carrier of remanence, whose direction is not significantly affected by the foliation. The reason for this, we propose, is that the fine-grained magnetite was produced by subsolidus exsolution from silicates *after* foliation was acquired. The directions of high coercive force, high T_{UB} components in **Y** specimens may have been affected by foliation, but there is no evidence that they have, and IRM experiments indicate that the effect, if present, is small. VRM's acquired in the laboratory during long-term heating experiments are small, one or two orders of magnitude less than NRM. For these reasons it is concluded that **X** and **Y** magnetization accurately record the paleofield during uplift and cooling, and that the integrated hornblende age (101.7 ± 0.7 Ma) is a good estimate of their age. Rocks with northerly directions (site 31; Fig. 6) have ages (Fig. 3) similar to those with northeasterly directions (site I8; Fig. 4). Hence, the elongate distribution is not the result of partial Cenozoic overprinting. The rate of cooling (about 60°/Ma) and the range of unblocking temperatures (600–500°C) indicate that specimens at each site probably integrate the field over sufficient time to average out some,

Fig. 14. (*a*) Site locations and strike ST of axis of rotation (Fig. 9). (*b*) Declination versus perpendicular distance to strike.



if not all, the paleosecular variation.

Z magnetizations generally have low and distributed T_{UB}'s, low coercive forces, and directions close to the present field. IRM acquisition and Lowrie–Fuller test show that the remanence carrier is the large, probably multidomain, magnetite visible in thin section. VRM acquired in long-term heating experiments is very large. For these reasons, **Z** magnetizations are considered to be Brunhes age VRM's.

Argillites from the Lower Cretaceous Gambier Group (I9 and I11, Table 1) have magnetizations carried by pyrrhotite and have directions close to the expected Tertiary field. Irving et al. (1985) attributed them to Tertiary thermal overprinting. However, geochronological results, described herein, provide no evidence of Tertiary reheating, and the argillite direcitons fall within the distribution of **X** and **Y** magnetizations. It is possible, therefore, that the argillite magnetization also was acquired, during uplift and cooling in mid-Cretaceous time.

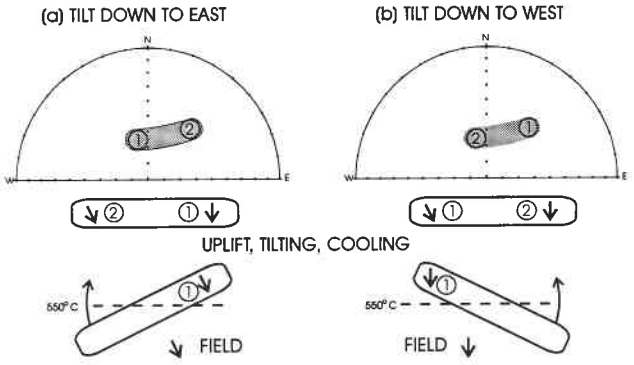
Tilt and latitudinal displacement

The elongated distribution of **X** magnetization and **Y** magnetizations freed of Brunhes overprints is consistent with tilting having occurred concurrent with uplift and cooling in mid-Cretaceous time—tilting about a horizontal axis, east of north (347°). The root mean square deviation of site mean directions from the small circle (the small circle is the intersection on a unit sphere of a cone with horizontal axis and semiangle of 64.4°) is 4.6° and the error is 2.2° ($P = 0.05$). No site direction differs significantly from this small circle, and the circle itself is significantly different from the expected North American mid-Cretaceous field.

In Fig. 14, declinations observed at sites on the east side of Howe Sound are plotted as a function of perpendicular distance from the strike line down through site 35. Declinations decrease towards the east, indicating that no major fault is present, and that the section was tilted as a single block. Declinations observed on the west side of Howe Sound do not fall on this line, indicating that minor fault(s) occur along the sound.

The data do not define the sense of tilt because the relative ages of magnetizations are not known; if tilt had been down towards the east (west) then directions of younger magnetizations would fall at the east (west) end of the distribution (Fig. 15). We identify three possibilities, represented by G, E, and F of Fig. 9.

Fig. 15. Models for acquisition of remanent magnetizations during uplift and cooling through blocking temperatures for tilting down to the east (*a*), and down to the west (*b*). Top: direction of Earth's field. Bottom: distribution of directions of magnetization produced. ①, magnetization direction acquired just before tilting commences; ②, magnetization direction acquired just after tilting stops.



G (D, I = 347.0°, 64.4°) is the point on the small circle with maximum inclination (64.4° ± 2.2° (P = 0.05)), which corresponds to a latitude of 44.5° ± 2.4°. The expected mid-Cretaceous latitude of the sampling locality (49.6°N, 123.2°W) is 60.5° ± 3.6°. The difference, 14.8° ± 3.9°, implies a latitudinal displacement of 1600 ± 400 km from the south since mid-Cretaceous time. In this interpretation, G has approximately the same declination as the expected paleofield, so no significant rotation about a vertical axis is required.

E (D, I = 354.2°, 62.5°, K = 240, a_{95} = 4.3°) is the average of the six westernmost site mean directions and is an estimate of position 2 for the hypothesis of tilt down to the west (Fig. 15). The inclination corresponds to a latitude of 43.8° ± 4.2°, which is 16.7° ± 5.5° less than expected, yielding a displacement from the south of 1900 ± 600 km.

F (D, I = 035.6°, 50.8°, K = 216, a_{95} = 4.6°) is the average of the six easternmost site mean directions and is an estimate of position 2 for the hypothesis of tilt down to the east. The inclination corresponds to a latitude of 31.5° ± 3.5°, which is 29.0 ± 4.9° less than the expected, yielding a displacement from the south of 3200 ± 500 km. This interpretation also requires a clockwise vertical-axis rotation of 60.5° ± 9.0°.

If E and F are good estimates of the limits of the distribution (Fig. 9) then the mid-Cretaceous tilt required is 29°. This is very much greater than the westward-down Cenozoic tilt (4–5°) implied by fission-track data obtained from the Coast Belt (Parrish 1983).

Discussion

The largest estimate of displacement (from F, above) compares favourably with the estimate of 3000 ± 600 km obtained from mid-Cretaceous sedimentary and igneous rocks of the eastern Coast Belt, in which the paleohorizontal

has been estimated from plane laminated siltstones (Wynne et al. 1995), and with the estimate of 3100 ± 600 km obtained for the Mt. Stuart batholith, also of the eastern Coast Belt, in which the paleohorizontal has been estimated by hornblende geobarometry (Beck et al. 1981; Ague and Brandon 1992). The largest estimate also compares with the displacement of 2500 ± 700 km proposed for the Spuzzum Pluton (SP, Fig. 1) assuming that it has not been substantially tilted since emplacement (Irving et al. 1985). From petrologic studies, Brown and Burmester (1991) argued that the pluton has been tilted first down to the northeast and then down to the southwest, so that its present attitude is close to that of the original. They identified an initial metamorphism at the time the pluton was intruded (94 Ma), and a later one spacially associated with the nearby Scuzzy Pluton (84 Ma). They proposed that the magnetization of Spuzzum dates not from its intrusion but from the second metamorphism, and thus has been affected by the second phase of tilting, which they argue, is sufficient to explain the anomalous magnetization without recourse to large-scale transport. In contrast, based on a review of numerous age studies, Irving et al. (1985) argued that the magnetization (observed mainly from the southern part of Spuzzum and distant from the Scuzzy Pluton with its associated younger metamorphism) was acquired following intrusion between 105 and 90 Ma. Thus the magnetization directions would have been affected by both phases of tilting, the sum of which approximates zero. Consequently, the assumption of little or no tilt may be essentially correct, and the displacement estimate of Irving et al. (1985) still valid. Further studies of Spuzzum are required, but the evidence at present is consistent with the largest estimate of displacement of the Porteau Pluton. Accurate comparisons with results from other mid-Cretaceous plutons of the Coast Belt farther north near Prince Rupert (102 and 109 Ma, Symons 1974, 1977*b*) and from the southern Coast Belt (Vandall 1993) are not possible because tilt estimates have not been made.

In summary, paleomagnetic data obtained from the Porteau Pluton, are fully consistent with tectonic models that require displacement of the Coast Belt of British Columbia from the south relative to cratonic North America by distances well in excess of 1500 km and perhaps by as much as 3000 km. They are inconsistent with models advocated by Price and Carmichael (1986), Butler et al. (1989), Vandall (1993), and Monger et al. (1994), which would limit post mid-Cretaceous displacements of the Coast Belt to 1000 km or less.

Acknowledgments

We are very pleased to have our paper included in this memorial to Richard Lambert. Two of the authors, E.I. and D.Y., who knew him from his undergraduate and early teaching days in Cambridge and Oxford, recall his wide-ranging interests and knowledge of all things geological, and remember him as a stimulating and argumentative friend.

We are grateful to D. Thorkelson for help with fieldwork, to P. Smith for help and guidance with operating the laser dating system, to T. Hamilton and P.J. Wynne for petrological discussions, to C. Lowe for providing aeromagnetic data, to L. Pearson, S. Friesen, and K. Light for typing the manuscript, and to L. Bedard and R. Franklin for drafting the diagrams. The manuscript was reviewed by M.E. Beck, Jr., R.F. Butler, H. Gabrielse, and P. Smith. The dating work was supported by Natural Sciences and Engineering Research Council of Canada funds.

References

Ague, J.J., and Brandon, M.T. 1992. Tilt and northward offset of Cordilleran batholiths resolved using igneous barometry. Nature (London), **360**: 146–149.

Beck, M.E. 1976. Discordant paleomagnetic pole positions as evidence of regional shear in the western Cordillera of North America. American Journal of Science, **276**: 694–912.

Beck, M.E. 1980. Paleomagnetic record of plate-margin tectonic processes along the western edge of North America. Journal of Geophysical Research, **85**: 7115–7131.

Beck, M.E. 1991. Some thermal and paleomagnetic consequences of tilting a batholith. Tectonics, **11**: 297–302.

Beck, M.E. 1992. Tectonic significance of paleomagnetic results for the western conterminous United States. *In* The Cordilleran Orogen: conterminous United States. *Edited by* B.C. Burchfield, P.W. Lipman, and M.L. Zoback. Geological Society of America, Geology of North America, Vol. G3, pp. 683–696.

Beck, M.E., and Noson, L. 1972. Anomalous paleolatitude in Cretaceous granitic rocks. Nature (London), **235**: 11–13.

Beck, M.E., Burmester, R.F., and Schoonover, R. 1981. Paleomagnetism and tectonics of Cretaceous Mt. Stuart batholith of Washington: translation or tilt? Earth and Planetary Science Letters, **567**: 336–342.

Brown, E.M., and Burmester, R.F. 1991. Metamorphic evidence for tilt of the Spuzzum pluton: diminished basis for Baja British Columbia concept. Tectonics, **10**: 978–985.

Butler, R.F., Gehrels, G.E., McClelland, W.C., May, S.R., and Klepacki, D. 1989. Discordant paleomagnetic poles from the Canadian Coast Plutonic Complex: regional tilt rather than large-scale displacement. Geology, **17**: 691–694.

Harrison, T.M. 1981. Diffusion of ^{40}Ar in hornblende. Contributions to Mineralogy and Petrology, **78**: 324–331.

Harrison, T.M., Duncan, I., and McDougall, I. 1985. Diffusion of ^{40}Ar in biotite: Temperature, pressure, and compositional effects. Geochimica et Cosmochimica Acta, **49**: 2461–2468.

Irving, E., and Archibald, D.A. 1990. Bathozonal tilt corrections to paleomagnetic data from mid-Cretaceous plutonic rocks: examples from the Omineca belt, British Columbia. Journal of Geophysical Research, **95B**: 4579–4585.

Irving, E., and Wynne, P.J. 1990. Paleomagnetic evidence bearing on the evolution of the Canadian Cordillera. Philosophical Transactions of the Royal Society of London, Series A, **331**: 487–509.

Irving, E., Woodsworth, G.J., Wynne, P.J., and Morrison, A. 1985. Paleomagnetic evidence for

displacement from the south of the Coast Plutonic Complex, British Columbia. Canadian Journal of Earth Sciences, **22**: 584–598.

Layer, P.W., Hall, C.M., and York, D. 1987. The derivation of $^{40}Ar/^{39}Ar$ age spectra of single grains of hornblende and biotite by laser step-heating. Geophysical Research Letters, **14**: 756–760.

Lowrie, W., and Fuller, M. 1971. On the alternating demagnetization characteristics of multidomain thermoremanent magnetization in magnetite. Journal of Geophysical Research, **76**: 6339–6349.

Lynch, J.V.C. 1991. Georgia Basin Project: stratigraphy and structure of Gambier Group rocks in the Howe Sound – Mamquam River area, southwest Coast Belt, British Columbia. *In* Current research, part A. Geological Survey of Canada, Paper 91-1A, pp. 49–57.

Monger, J.W.H., van der Heyden, P., Journeay, J.M., Evenchick, C.A., and Mahoney, J.B. 1994. Juro-Cretaceous basins along the Canadian Coast Belt: their bearing on pre-mid-Cretaceous sinistral displacement. Geology, **22**: 175–178.

Parrish, R.R. 1983. Cenozoic thermal evolution and tectonics of the Coast Mountains of British Columbia 1. Fission track dating, apparent uplift rates, and patterns of uplift. Tectonics, **2**: 601–631.

Price, R.,A., and Carmichael, D.M. 1986. Geometric test for Late Cretaceous – Paleogene intracontinental transform faulting in the Canadian Cordillera. Geology, **14**: 468–471.

Roddick, J.A. 1987. Day 4, Lillooet to Vancouver via Duffey Lake, Coast Plutonic Complex and (?) Cascade Fold Belt. *In* Cordilleran cross-section: Calgary to Vancouver. *Edited by* R.A. Price, R.R. Parrish, J.W.H. Monger, and J.A. Roddick. *Edited by* 19th General Assembly of the International Union of Geodesy and Geophysics, Guidebook for Scientific Excursion Al, pp. 58–85.

Roddick, J.A., and Woodsworth, G.J. 1979. Geology of Vancouver west half and mainland part of Alberni. Geological Survey of Canada, Open File 611.

Roddick, J.C.M. 1983. High precision intercalibration of $^{40}Ar/^{39}Ar$ standards. Geochimica et Cosmochimica Acta, **47**: 887–898.

Symons, D.T.A. 1973. Concordant Cretaceous paleolatitudes from felsic plutons in the Canadian Cordillera. Nature (London), **241**: 59–61.

Symons, D.T.A. 1974. Age and tectonic implications of paleomagnetic results from plutons near Prince Rupert, British Columbia. Journal of Geophysical Research, **79**: 2690–2697.

Symons, D.T.A. 1977*a*. Geotectonics of Cretaceous and Eocene plutons in British Columbia: a paleomagnetic fold test. Canadian Journal of Earth Sciences, **14**: 1246–1262.

Symons, D.T.A. 1977*b*. Paleomagnetism of Mesozoic plutons in the westernmost Coast Complex, British Columbia. Canadian Journal of Earth Sciences, **14**: 2127–2139.

Turner, G., Hunkeke, J., Podosek, F.A., and Wasserburg, G.J. 1971. $^{40}Ar-^{39}Ar$ ages and cosmic ray exposure ages of Apollo 14 samples. Earth and Planetary Science Letters, **12**: 19–35.

Vandall, T.A. 1993. Cretaceous Coast Belt paleomagnetic data from the Spetch Creek pluton, British Columbia: evidence for the "tilt and moderate displacement" model. Canadian Journal of Earth Sciences, **30**: 1037–1048.

van Fossen, J., and Kent, D. 1992. Paleomagnetism of 122 pluton in New England and the mid-Cretaceous paleomagnetic field in North America; true polar wonder or large-scale differential mantle motion. Journal of Geophysical Research, **97B**: 19 651 – 19 661.

Wright, N., Layer, P.W., and York, D. 1990. New insights into thermal history from single grain $^{40}Ar/^{39}Ar$ analysis of biotite. Earth and Planetary Science Letter, **104**: 70–79.

Wynne, P.J., Irving, E., Maxson, J., and Kleinspehn, K. 1995. Paleomagnetic results from the middle Cretaceous Silverquick and Powell Creek formations: evidence for 3000 km northward displacement of the eastern Coast Belt, British Columbia, Journal of Geophysical Research, **99B**, in press.

York, D. 1984. Cooling histories from $^{40}Ar/^{39}Ar$ spectra: implications for Precambrian plate tectonics. Annual Review of Earth and Planetary Sciences, **12**: 383–409.

Zartman, R.E. 1964. A geochronological study of the Lone Cove pluton from the Llano uplift, Texas. Journal of Petrology, **5**: 359–408.